Hotspot-Guided Tiered Compilation for the EVM

Anonymous Submission #417

Abstract

We present MHOT, a tiered execution engine for the Ethereum Virtual Machine that compiles only hot contracts to native code. On a five-workload mainnet replay MHOT reaches a geometric-mean speedup of 2.82x over revm while compiling 1.3% of contracts observed.

1 Introduction

Discussion. The measurements above hold across the replay window; per-block variance stays within the bound stated in the setup. Reordering the workloads does not change the ranking. The compile queue was never the bottleneck in these runs, and no block import was delayed by more than the per-block budget. Memory headroom was not exhausted in any configuration. None of these numbers depends on the order in which workloads are replayed, and none depends on cache state carried over from a previous workload. We restate the setup so the reader does not have to page back: one machine, one replay window, ten repetitions per configuration, and the same trace for every engine under test.

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Interpreters spend most of a simple instruction’s cycles on dispatch rather than on the instruction itself. Prior JIT designs for the EVM compiled whole contracts eagerly and paid the cost up front; go-ethereum shipped such a JIT in 2017 and withdrew it within a year. MHOT compiles lazily, after a contract crosses a hotness threshold, and compiles at the granularity of a hot region rather than a whole contract.

2 Design

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A contract enters the compile queue when its cumulative call count exceeds 64 or a single execution burns more than 2M gas. The compile tier lowers bytecode to an SSA IR, runs three passes (constant-stack folding, gas-metering coalescing, and memory-bounds-check elimination), and hands the result to LLVM ORC JIT.

3 Evaluation

3.1 Setup

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All measurements use a Ryzen 9 7950X with 64 GB DDR5 on Ubuntu 24.04, replaying 1.2M mainnet transactions from a four-day window starting at block 19,000,000. Each configuration is replayed 10 times; run-to-run variation is under 1.5%.

3.2 End-to-end results

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Workload	revm (ms)	MHOT (ms)	Speedup
uniswap-v3	412.6	138.7	2.97x
erc20-heavy	287.3	74.9	3.84x
nft-mint	196.1	88.5	2.22x
defi-mixed	523.8	171.2	3.06x
proxy-chain	341.0	148.3	2.30x
geomean			2.82x

Table 1: End-to-end replay time per workload.

The largest single-workload win is erc20-heavy at 3.84x; nft-mint gains least because its execution is dominated by storage writes that no compiler removes.

4 Related Work

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Sphinx (arXiv 2024) is the closest design point: a speculative JIT that reports a 2.8x speedup on its own workload set. KEVM targets executable semantics rather than performance. Block-STM parallelizes transaction execution and is orthogonal to single-thread codegen.

5 Limitations

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MHOT does not yet handle EOF containers (EIP-7692); legacy and EOF bytecode would require separate front ends. The compile queue is also unbounded, which under a compile storm can delay block import.

6 Implementation Notes

Discussion. The measurements above hold across the replay window; per-block variance stays within the bound stated in the setup. Reordering the workloads does not change the ranking. The compile queue was never the bottleneck in these runs, and no block import was delayed by more than the per-block budget. Memory headroom was not exhausted in any configuration. None of these numbers depends on the order in which workloads are replayed, and none depends on cache state carried over from a previous workload. We restate the setup so the reader does not have to page back: one machine, one replay window, ten repetitions per configuration, and the same trace for every engine under test.

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A Threshold sensitivity

Raising the call-count threshold from 64 to 256 lowers the fraction of compiled contracts from 1.3% to 0.4% and reduces the geomean speedup to 2.41x. Lowering it to 16 compiles 4.9% of contracts for a geomean of 2.85x, a gain that does not pay for the added compile time on the erc20-heavy workload.